Elementary Statistics for Social Sciences II

208272 (Part 1)

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Preface

This book covers the first half of 208272 Elementary Statistics for Social Sciences II: the principles of statistical inference, inference about means, proportions and variances, and analysis of variance. All examples use economic contexts. Unless a source is given, the numbers in the examples are hypothetical and are used for illustration only.

How to use this book

Each section of Chapters 2–4 matches one class session and follows the same pattern:

1. **Learning objectives**
2. **Concept**: why we need this method
3. **Formulas and conditions**
4. **Worked example**: a full solution
5. **Your Turn**: a similar problem with space to write your own solution
6. **Reading R output** (from Section 3.2 onward)
7. **Formula summary**
8. **Exercises**
9. **Solutions** to Your Turn and the Exercises

In the PDF, the Your Turn problems have blank space for writing, and the solutions are at the end of each section. On the web, click a solution box to open it. Try the problem first.

Session	Section
1	Chapter 1 Basic Principles of Statistical Inference
2	2.1 Mean, σ known
3	2.2 Mean, σ unknown
4	2.3 Proportion
5	2.4 Variance
6	3.1 Two proportions
7	3.2 P-values and R output; 3.3 Paired means
8	3.4 Two variances
9	3.5 Two independent means
10	4.1 One-way ANOVA: CRD
11	4.2 Multiple comparisons (Tukey); 4.3 RCBD
12	Summary: Choosing a Method

Notation for critical values

The subscript of a critical value is the **area to its left**. This is the same convention as the R functions `qnorm()`, `qt()`, `qchisq()` and `qf()`.

Notation	Meaning	R
$z_{0.975}$	the z value with area 0.975 to its left	`qnorm(0.975)`
$t_{0.975,\,24}$	the t value with 24 df and area 0.975 to its left	`qt(0.975, 24)`
$\chi^2_{0.025,\,10}$	the χ^2 value with 10 df and area 0.025 to its left	`qchisq(0.025, 10)`
$F_{0.975,\,5,\,8}$	the F value with (5, 8) df and area 0.975 to its left	`qf(0.975, 5, 8)`

A value calculated from the sample has the subscript "cal", for example z_{cal}, t_{cal}, χ^2_{cal} and F_{cal}.

The five steps of a hypothesis test

1. **Hypotheses**: state H_0 and H_1.
2. **Level of significance**: state α.
3. **Test statistic**: write the formula, substitute the values, and give the result.
4. **Critical value and critical region**.
5. **Decision and interpretation**.

Writing the conclusion

Decision using the critical region (Chapters 1–2 and Section 3.1)

> t_{cal} = … [falls / does not fall] in the critical region (t_{cal} [<, >] …), so we [reject / do not reject] H_0.

Decision using the p-value (from Section 3.2 onward)

> p-value = … [<, ≥] $\alpha = 0.05$, so we [reject / do not reject] H_0.

Interpretation

> That is, … at the 0.05 level of significance.

- If we reject H_0, write H_1 as a sentence.

- If we do not reject H_0, write H_0 as a sentence.
- For a two-sided test that rejects H_0, also state the direction, using the sample statistic.

Confidence interval

> We are 95% confident that … is between … and ….

Rounding

To be added.

1 Basic Principles of Statistical Inference

Learning Objectives

To be added.

1.1 Sampling Distributions of $\bar{x}$ and $\hat{p}$

To be added.

1.2 Standard Error and the Central Limit Theorem

To be added.

1.3 The Idea of a Confidence Interval

To be added.

1.4 The Logic of Hypothesis Testing

To be added.

1.5 Type I and Type II Errors

To be added.

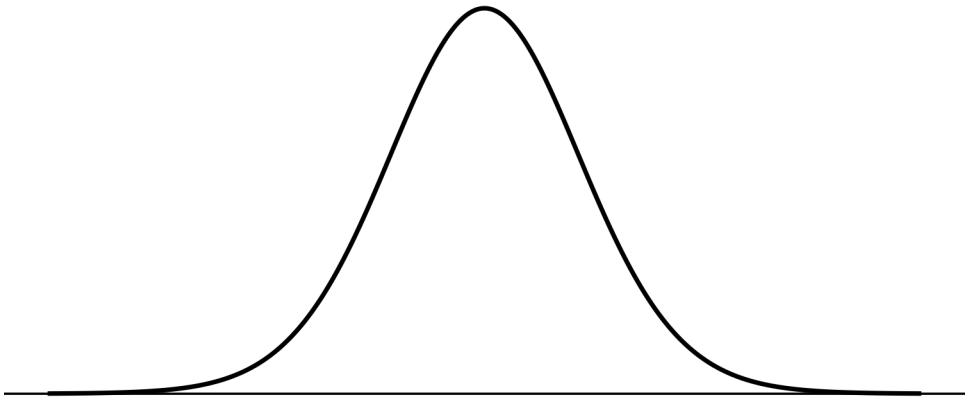


Figure 1.1: Blank curve: shade the area to the left of $t_{0.975,\,24}$ (layout demo)

1.6 Reading Statistical Tables

To be added. Shade the area to the left of the value before you read the table.

Formula Summary

To be added.

Exercises

To be added.

2 One-Sample Inference

2.1 Mean, σ Known

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.1.1

Problem to be added.

Step 1: Hypotheses

H_0: _____

H_1: _____

Step 2: Level of significance

$\alpha =$ _____

Step 3: Test statistic

Formula: _____

Substitution: _____

Result: _____

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.1.1

Solution to be added.

2.2 Mean, σ Unknown

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.2.1

Problem to be added.

Step 1: Hypotheses

H_0: _____

H_1: _____

Step 2: Level of significance

$\alpha =$ _____

Step 3: Test statistic

Formula: _____

Substitution: _____

Result: _____

Step 4: Critical value and critical region

Critical value(s): _____

Critical region: _____

Step 5: Decision and interpretation

Decision: _____

_____

Interpretation: _____

_____

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.2.1

Solution to be added.

2.3 Proportion

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.3.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.3.1

Solution to be added.

2.4 Variance

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 2.4.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 2.4.1

Solution to be added.

3 Two-Sample Inference

3.1 Two Proportions

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.1.1

Problem to be added.

Step 1: Hypotheses

Step 2: Level of significance

Step 3: Test statistic

Step 4: Critical value and critical region

Step 5: Decision and interpretation

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.1.1

Solution to be added.

3.2 P-values and R Output

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.2.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.2.1

Solution to be added.

3.3 Paired Means

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.3.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.3.1

Solution to be added.

3.4 Two Variances

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.4.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.4.1

Solution to be added.

3.5 Two Independent Means

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 3.5.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 3.5.1

Solution to be added.

4 Analysis of Variance

4.1 One-Way ANOVA: CRD

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.1.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.1.1

Solution to be added.

4.2 Multiple Comparisons (Tukey)

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.2.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.2.1

Solution to be added.

4.3 Randomized Complete Block Design (RCBD)

Learning Objectives

To be added.

Concept

To be added.

Formulas and Conditions

To be added.

Worked Example

To be added.

Your Turn

Your Turn 4.3.1

Problem to be added.

Reading R Output

To be added.

Formula Summary

To be added.

Exercises

To be added.

Solutions

Solution: Your Turn 4.3.1

Solution to be added.

Summary: Choosing a Method

Which Method? A Flowchart

To be added.

Mixed Problems

To be added.

Writing a Full Conclusion

To be added.

A Statistical Tables

All tables give the value with the stated **area to its left**.

A.1 Standard Normal Distribution

To be added.

A.2 t Distribution

To be added.

A.3 Chi-Square Distribution

To be added.

A.4 F Distribution

To be added.